

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.

**This answer is to be made in separate loose script(s) provided for the purpose.
Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.**

**On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.**

Answer any five questions from Group-A, B & C, taking at least one from each group.

1. Answer the following questions (any twenty): 1x20
 - i) In Oxyacetylene gas welding with neutral flame, the ratio of acetylene to oxygen is – (a) 1:2, (b) 1:1, (c) 2:1, (d) 1:3.
 - ii) Non-consumable electrode is used in the – (a) MIG welding, (b) TIG Welding, (c) Arc welding, (d) Gas welding.
 - iii) Chances of crack propagation are more in – (a) cold working process, (b) hot working process, (c) both (a) & (b), (d) none of these.
 - iv) The gating ratio 2:8:1 for copper in gating system design refers to the ratio of areas of – (a) sprue : runner : ingate, (b) runner: ingate : sprue, (c) runner : sprue : ingate, (d) ingate : runner : sprue.
 - v) The centre lathes receive their power through – (a) headstock, (b) tailstock, (c) both (a) & (b), (d) none of these.
 - vi) In oblique cutting system, the maximum chip thickness – (a) occurs at the middle, (b) may not occur at the middle, (c) depends upon the material of the tool, (d) depends upon the geometry of the tool.
 - vii) Which current is used in Tungsten Inert-Gas (TIG) welding? – (a) Only A.C. can be used as welding current, (b) Only D.C. can be used as welding current, (c) Both A.C. and D.C. can be used as welding current, (d) None of these.
 - viii) Gate is provided in moulds to – (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - ix) A moving mandrel is used in – (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) forging.
 - x) Scab is a – (a) welding defect, (b) sand casting defect, (c) machining defect, (d) forging defect.
 - xi) Which of the following gas mixtures is not used in Gas Tungsten arc welding (TIG)? – (a) Argon-Helium, (b) Argon-Nitrogen, (c) Argon-Hydrogen, (d) Argon-CO₂.
 - xii) Which of the following is always a cold working operation? – (a) Rolling, (b) Wire drawing, (c) Forging, (d) Extrusion.
 - xiii) The angle between face and flank of a single point cutting tool is known as _____.
 - xiv) In _____ welding process, the arc is not visible to worker.

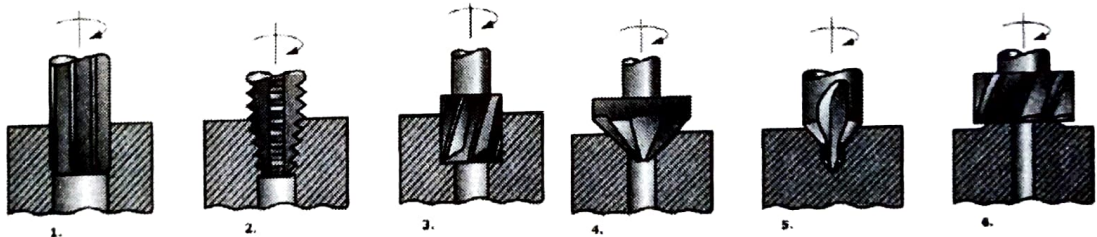
- xv) _____ is a mixture of finely divided aluminum and iron oxides.
- xvi) Cuts and washes in sand casting is caused by _____ of molding sand by the flowing metal.
- xvii) Spot/Seam/TIG welding is used in automobile industry.
- xviii) Hot working of metal in metal forming processes is working _____ recrystallization temperature.
- xix) In MIG welding, _____ current _____ polarity is generally used.
- xx) Single point cutting tool should ideally have _____ rake angle.
- xxi) A lathe bed is generally made of _____.

Group-A

- 2. a) What do you mean by forming? What is recrystallization?
 b) Distinguish between forward extrusion and backward extrusion with neat sketches. (1+2)+4+3
 c) What are the advantages of hot working over cold working process?
- 3. a) What do you mean by rolling? Discuss different kinds of roll stand arrangement.
 b) State three forging defects and state the reason for arising these defects. (1+4)+2+3
- 4. a) Discuss different stages of drop forging.
 b) What is the difference between punching and blanking? What is meant by stripping force and punching force? 5+5

Group-B

- 5. a) Differentiate between orthogonal cutting and oblique cutting.
 b) Explain the nomenclature of a single point cutting tool with a neat sketch. 4+6
- 6. a) Differentiate between Capstan and Turret lathe.
 b) Identify the following machining operations and define them. 4+6



- 7. a) How do you specify a milling machine?
 b) Differentiate between conventional milling & climb milling.
 c) Write short notes on any two – (a) Knurling, (b) facing, (c) Parting off. 2+4+4
- 8. a) What are the different types of chips formed during machining? Write short note on Continuous chip with built-up edges.
 b) Draw the neat sketch of a twisted drill bit and mark various part of it. (1+3)+6

Group-C

- 9. a) Define core print. What is the use of it?
 b) Write down the advantages and disadvantages of green sand mould.
 c) Write short notes on – (i) Sweep pattern, (ii) Loose-piece pattern. 2+4+4

10. a) Draw the diagram depicting two-piece sand casting arrangement and show the locations of pouring basin, sprue, gate, runner, riser, mould cavity, parting line, air vent, cope and drag on it.
b) Name three welding defects. State the reason for arising any one of such defects and the remedies to avoid them. 5+(1+2+2)
11. a) What are the typical electrode materials used for Tungsten Inert gas welding?
b) Explain the characteristics of Arc welding machines, which viz., constant current and constant voltage. Mention the applications of each of them.
c) What are the problems encountered with the use of coated electrodes? Explain how these can be taken care of. 2+4+4
12. a) Distinguish between – (i) hot chamber and cold chamber die casting, (ii) Soldering & Brazing.
b) Explain precision investment casting process with neat sketch. (3+3)+4
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